

CORAL REEFS

Heat-driven functional extinction of Caribbean *Acropora* corals from Florida's Coral Reef

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In 2023, a record-setting marine heat wave triggered the ninth mass coral bleaching event on Florida's Coral Reef (FCR). We examined spatial patterns of heat exposure along the ~560-kilometer length of FCR and the mortality of two ecologically important, critically endangered reef-building corals. Sea surface temperatures were $\geq 31^\circ\text{C}$ for an average of 40.7 days, leading to heat exposures 2.2- to fourfold higher than all prior years on record. In the Florida Keys and Dry Tortugas, 97.8 to 100% of the *Acropora palmata* and *Acropora cervicornis* colonies died. Mortality was lower offshore southeast Florida (37.9%), reflecting cooler temperatures in this region. Since the late 1970s, multiple stressors had already reduced the ecological relevance of *Acropora* in Florida, but the 2023 heat wave marks their functional extinction from FCR.

The frequency and severity of extreme climate and weather events, including marine heat waves, are increasing, resulting in widespread degradation of the function, structure, resilience, and adaptive capacity of ecosystems (1–3). Coral reefs are one of the most heat-vulnerable marine ecosystems; heat-driven mass bleaching events have caused mortality and reef degradation for >40 years, with impacts continually increasing in scale and severity (2, 3). Coral bleaching (the breakdown in the symbiosis between a coral and its algal endosymbionts) occurs when local temperatures are $\geq 1^\circ$ to 2°C than the maximum monthly mean temperature for a month or longer (4, 5). The risk of mortality is a function of the duration and magnitude of exposure, although extreme thermal anomalies ($+3^\circ$ to 4°C) can elicit mortality in a few days (4). Corals can recover from minor heat stress events but often endure years of impaired growth and reproduction as well as increased disease susceptibility (6–8).

Half of the Atlantic Ocean's reef-building coral species face a risk of extinction due to ocean warming and other threats in the coming two decades, with elkhorn (*Acropora palmata*) and staghorn coral (*Acropora cervicornis*) among the most critically endangered (9). In the absence of large-scale, ubiquitous stressors or catastrophic events, a global extinction (complete species disappearance) is often preceded by functional extinctions or extirpations at local (i.e., individual reefs) and regional scales. A functional extinction occurs when a species' abundance becomes so low that it no longer plays a major role in its ecosystem, such that there are resultant impacts to community structure and function (10). Extirpation (local or regional extinction) occurs when a species is displaced and no longer present within a defined location or region, shrinking the species distribution to a smaller realized range (10).

Heat stress and coral loss

We examined the spatial patterns of heat exposure along Florida's Coral Reef (FCR) and consequent mortality of *Acropora* spp. during the 2023 marine heat wave. The 2023 event was the ninth mass bleaching event to have affected FCR; previous events occurred in 1987, 1990, 1997, 1998, 2005, 2014, 2015, and 2022 (11) (see supplementary text). Beginning in early July, sea surface temperature (SST) quickly increased across FCR to the highest values ever recorded in the satellite record (Fig. 1). The highest recorded SSTs occurred in the Middle Florida Keys, with the regional mean SST peaking at 32.3°C on 15 July 2023 (table S1). The thermal anomalies, measured as hotspots ($^\circ\text{C}$: daily deviations above the maximum monthly mean SST) (5) were greatest in the Middle and Lower Florida Keys ($+2.5$ to 2.6°C), reaching values that were record setting by $+0.9^\circ\text{C}$ (table S1).

In this study, we document losses to both wild and restored populations of the two *Acropora* species along FCR during the 2023 marine heat wave (Fig. 2). As this event unfolded, in-water diver surveys ($n = 525$) tracked the fate of 52,356 wild and restored colonies across 391 distinct sites (table S2). The observed mortality matched the spatial pattern of accumulated heat stress (Fig. 3). Linear mixed-effects models showed that mortality ranged from 97.8 to 100% for both *Acropora* species in the Florida Keys and Dry Tortugas (fig. S1). Heat exposures in 2023 were unprecedented for the entire FCR, as measured by satellite-derived degree heating weeks (DHWs; $^\circ\text{C}\cdot\text{weeks}$) (Fig. 3), a metric used to quantify the duration and magnitude of accumulated heat stress (5). Maximum DHWs at the surveyed sites were on average threefold higher in 2023 compared with the 1986–2022 maxima (table S3). Examination of the Hadley Centre's long-term, interpolated SST dataset showed that 2023 was the hottest July–August period in 150+ years for FCR (fig. S2).

The magnitude of heat exposure was highest in the Middle and Lower Florida Keys, where all sites experienced 16.1 to $20.4^\circ\text{C}\cdot\text{weeks}$, with mean DHWs ($\pm$ standard error) equal to $18.2 (\pm 0.55)$ and $18.1 (\pm 0.16)^\circ\text{C}\cdot\text{weeks}$, respectively (table S3), resulting in nearly 100% mortality. Heat exposure declined north of the Florida Keys and, correspondingly, mean mortality declined from 90.8% in the Biscayne Region to 37.9% in Broward-Miami (table S2). These same patterns were apparent in the number of days that SST values exceeded threshold values that correlated with previous mass bleaching events on FCR (table S4). For example, prior research showed that monthly mean SSTs $\geq 30.4^\circ$ to 30.5°C were threshold values associated with the mass bleaching events in 1997, 1998, and 2005 (11). All sites experienced SSTs $\geq 30.4^\circ\text{C}$ for roughly 2 to 3 months in 2023, which shows that heat stress accumulation persisted for two- to threefold longer than during

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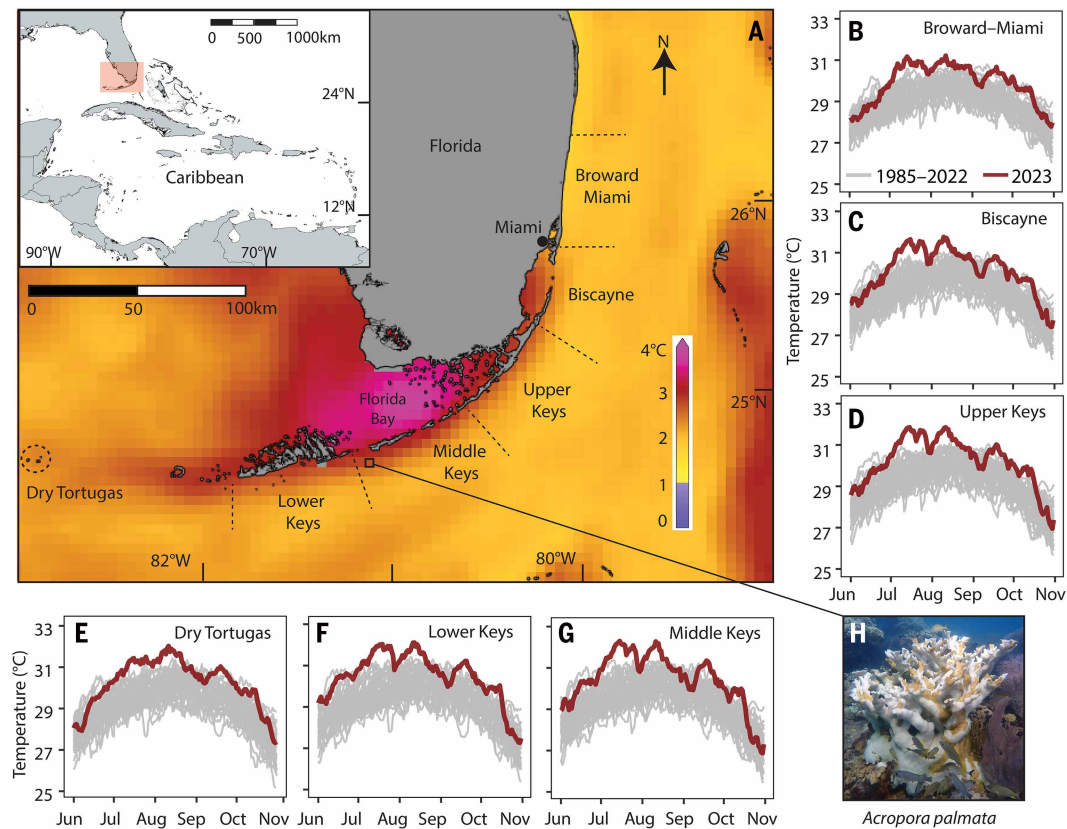


Fig. 1. Sea surface temperature patterns on Florida's Coral Reef in 2023. (A) Maximum hotspots (°C) in 2023. (B to G) Daily average sea surface temperature from 1985 to 2022 (gray) and 2023 (red) for six subregions of Florida's Coral Reef. (H) Colony of *Acropora palmata* showing rapid mortality and tissue sloughing at Sombrero Reef, 20 July 2023. Yellow-brown patches on colony are locations of tissue loss. [Photo credit, (H): Coral Restoration Foundation]

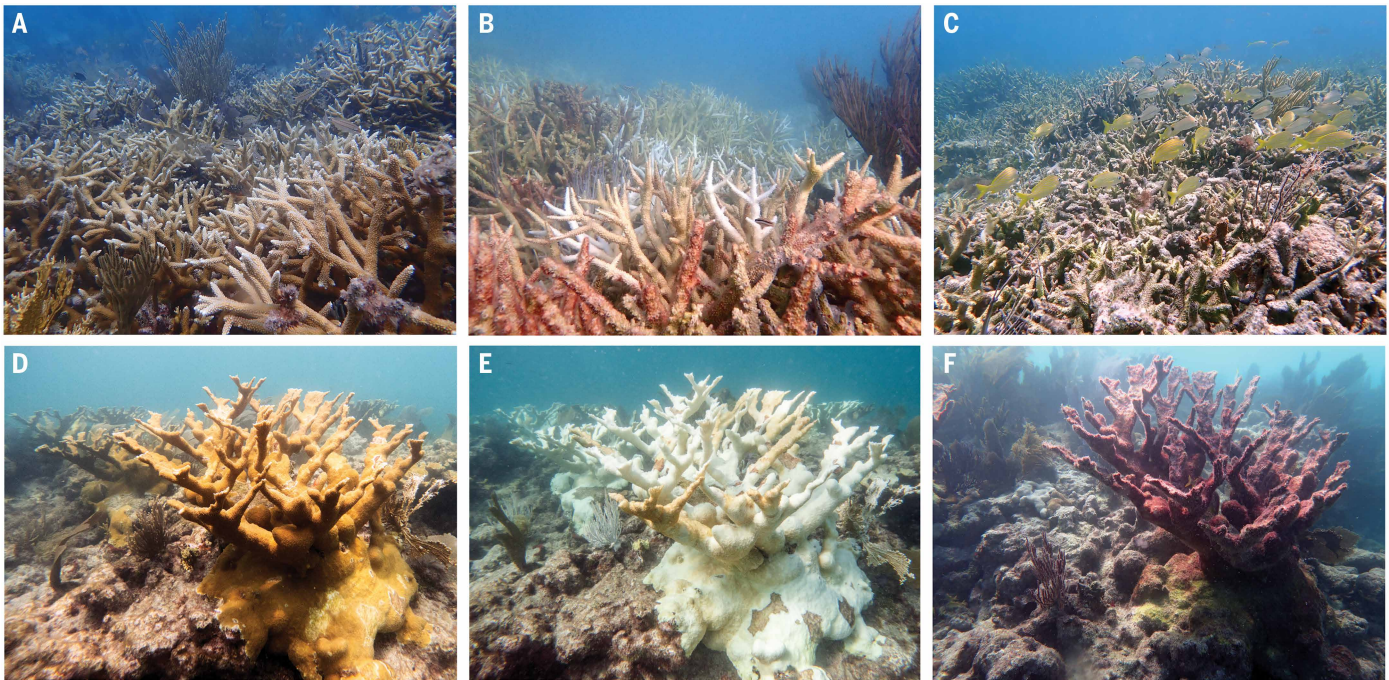


Fig. 2. Bleaching and mortality of *Acropora* on Florida's Coral Reef. Wild *Acropora cervicornis* thicket in Dry Tortugas: (A) non-bleached, 23 June 2023; (B) mostly dead, 11 September 2023; and (C) dead, 9 September 2023. Wild *A. palmata* colonies in Key Largo, Upper Florida Keys: (D) pale, 21 July 2023; (E) bleached, 18 August 2023; and (F) dead, 27 November 2023. [Photo credits: (A to C), Shedd Aquarium/S. Matsuda; (D and E), O. Williamson; (F) D. E. Williams]

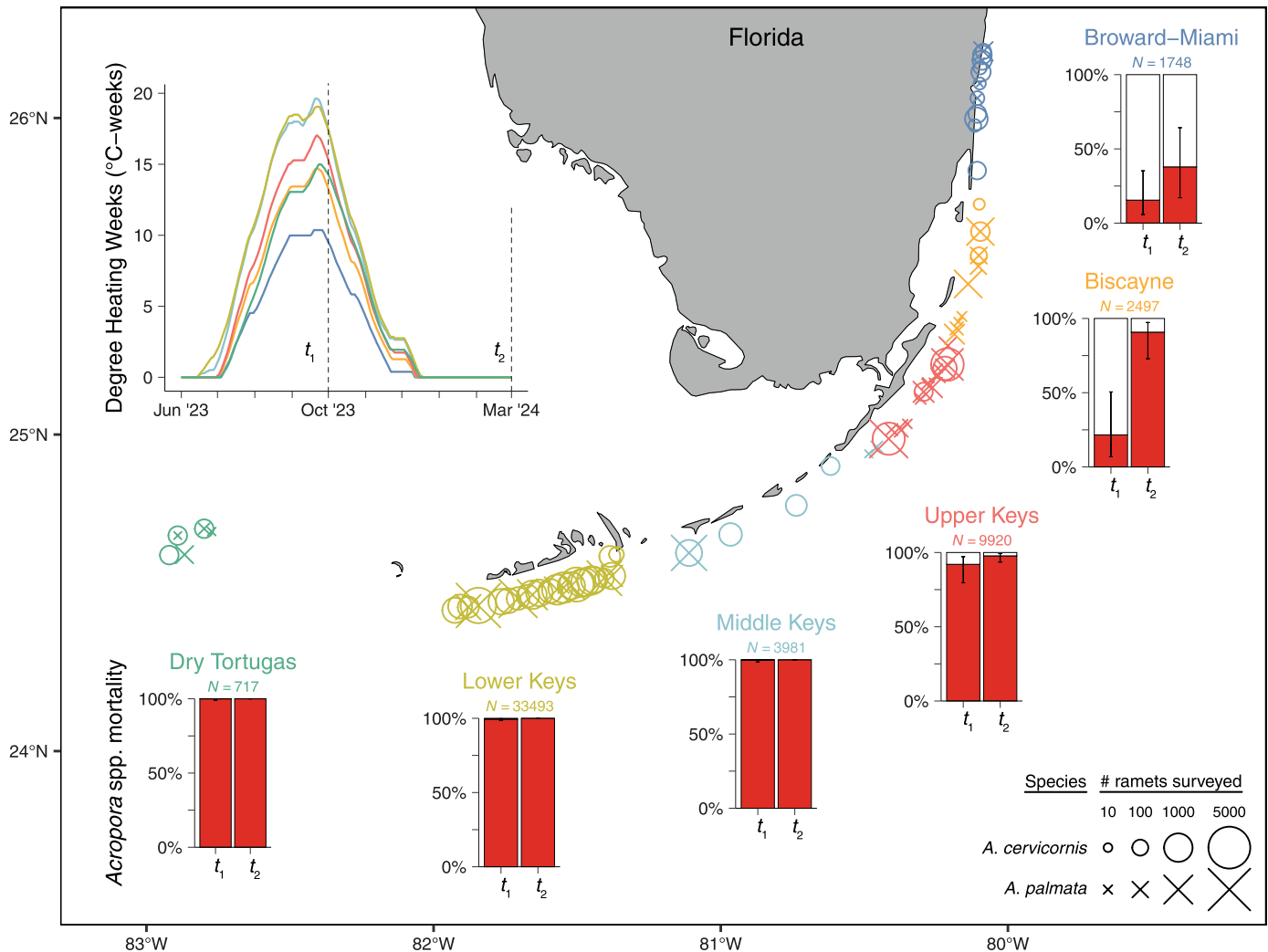


Fig. 3. Overall mortality of *Acropora* spp. on Florida's Coral Reef. Points represent site clusters where *A. palmata* (X's) and *A. cervicornis* (circles) were surveyed, with point size scaled by the number of distinct ramets observed ("ramet" refers to an individual, physically separate coral colony). Bar plots show the percent mortality ($\pm 95\%$ confidence interval) of all *Acropora* (which did not differ between species) on 1 October 2023 (t_1) and 1 March 2024 (t_2), with the conservative estimates of the total number of ramets observed in each subregion given above the bar plots. (Inset) Mean degree heating week trajectory colored by subregion, with dashed vertical lines at the two time points of interest.

the 1997, 1998, and 2005 events (table S4). It was also found that 1 day $\geq 31^\circ\text{C}$ occurred during past mass bleaching at Molasses Reef in the Upper Keys, but daily average SSTs did not reach this value in non-bleaching years (11). In 2023, SSTs $\geq 31^\circ\text{C}$ were sustained for 25 to 61 days across all sites from the Biscayne region southwest to the Dry Tortugas (table S4). SSTs were $\geq 31^\circ\text{C}$ for 4 to 21 days in the Broward-Miami subregion, further showing the lower heat stress in that region.

In the Middle and Lower Florida Keys, acute thermal stress led to tissue sloughing and rapid coral mortality in a matter of days (Fig. 1H). This represents acute heat shock, as opposed to the longer-term starvation or subsequent disease, competitive overgrowth, or predation that occurs when a coral loses and fails to regain its algal symbionts over weeks to months (4). A similar response occurred in *Acropora* on the northern Great Barrier Reef during bleaching in 2016, where extreme hotspot values near or above $+2^\circ\text{C}$ also occurred (3). This suggests that heat-sensitive *Acropora* spp. in the Indo-Pacific and Caribbean have an upper thermal limit near hotspot values of $+2^\circ\text{C}$, whereby mortality occurs almost immediately, irrespective of longer-term heat exposure accumulation. It is notable that nighttime satellite SSTs are used to avoid biases due to solar heating of the ocean skin interface (5), but they do not capture midday thermal

peaks and may underestimate acute heat stress, especially in shallow water. This pattern of acute heat shock and rapid coral mortality is an underappreciated, emerging threat to coral reefs that is becoming more common as extreme thermal anomalies increase in frequency and severity.

During summer in the Florida Keys, in situ subsurface temperatures are often warmer than those measured from satellite-based SST, and this pattern is most pronounced in the Middle and Lower Florida Keys (11). Dense, salty, and hot waters exit shallow Florida Bay through the wide channels in this region, pass over the reefs offshore, and have been implicated in the decline of the Florida Keys reefs (12). This leads to an inverted thermocline, whereby bottom temperatures are higher than those at the surface. At a coral nursery off Marathon in the Middle Keys on 12 and 14 July 2023, daily average in situ temperatures at 7.3 m peaked at 33.3°C , which was 1.0° and 0.9°C greater than the SST values measured on those days, respectively (fig. S3). Hourly values at this site reached 34.0°C on 13 July and were as much as 1.65°C greater at 7.3 m than 3 m (fig. S3). Some of the most dramatic examples of acute heat shock and rapid tissue loss were observed in this region 1 week later on 20 July (Fig. 1H). The presence of warmer bottom temperatures may be exacerbating ocean warming impacts in the Florida Keys, but the mechanism creating these subsurface

water masses, and their footprint, remains poorly understood and needs further study.

Modeled mortality response curves showed that *A. palmata* was more heat-sensitive than *A. cervicornis*, but because of intercolony variability, this trend was not significant (Fig. 4). Mortality began for *A. palmata* around 3 °C-weeks, whereas *A. cervicornis* did not show mortality until >4 °C-weeks (Fig. 4). *A. palmata* and *A. cervicornis* experienced 50% mortality at 7.8 and 10.8 °C-weeks, respectively, but the thermal stress that led to 95% mortality was similar for both species (17.6° and 17.7 °C-weeks, respectively). Mortality occurred at lower heat exposures and, thus, earlier during the 2023 heatwave for *A. palmata*, suggesting that future interventions during bleaching (e.g., coral rescues and relocations, among others) can justifiably prioritize *A. palmata*.

The northerly reaches of FCR were less affected by the 2023 marine heat wave. Indeed, nearly all of the *Acropora* that were still alive in March 2024 were located in the Biscayne and Broward-Miami subregions (Fig. 3 and table S2). FCR consists of two regions that are characterized by distinct geologic histories and geomorphology (13). The Florida Keys reef tract (FKRT) extends roughly 350 km from Miami southwest to the Dry Tortugas; this region contains topographically complex reef structures and has recently supported abundant populations of reef-building corals (13–16). Conversely, the southeast Florida continental reef tract (SFCRT) extends ~200 km north from Miami and consists of relict, fossil reef structures with low topographic complexity and low coral cover (13, 17). By 5800 years ago, active reef accretion no longer occurred on the SFCRT, and by 3000 years ago, most reefs in the FKRT were geologically senescent (13). This contraction of reef growth to lower latitudes is attributed to climatic cooling leading to more frequent and severe cold fronts (13). Despite this, there were notable populations of both *Acropora* species and ample records of reefs with abundant coral cover until the mid-1970s from the FKRT, indicating that many coral reef ecosystem functions persisted until ~40 to 50 years ago (14–16, 18, 19).

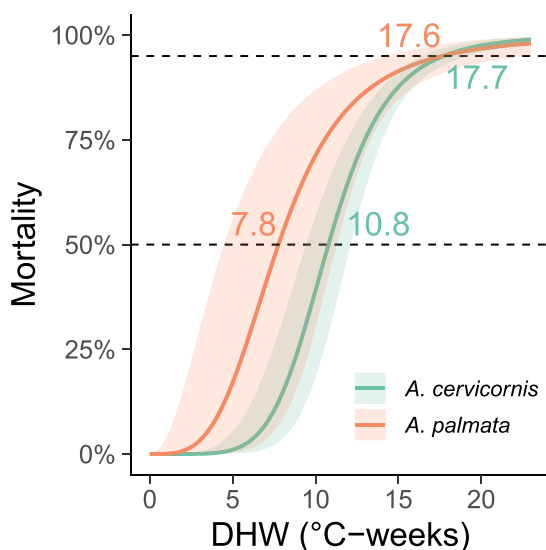


Fig. 4. Mortality response curves for *Acropora* spp. with increasing heat stress. Dose-response curves show the level of mortality observed at given levels of heat stress accumulation (recorded on or before each survey date) for both *Acropora* species, and shading represents bootstrapped 95% prediction intervals. Horizontal dashed lines indicate 50 and 95% mortality, respectively, and the corresponding values of degree heating weeks (DHWs) required to reach those effective dose (ED) thresholds (ED50 and ED95, respectively) in each species are annotated. There were no significant differences in ED50 or ED95 between species.

An uncertain future for Caribbean acroporids

Caribbean *Acropora* spp. have been in peril for at least the past 40 to 50 years; roughly 80% of the remaining populations were lost from the 1980s to 2012 (20). Florida's populations have likewise declined since the late 1970s [initially from disease (21)] and had failed to recover before the 2023 marine heat wave (22–25). The initiation of sustained, long-term monitoring of Florida Keys reefs in 1996 clearly illustrates the decline of these species and the deleterious impacts of bleaching (fig. S4). These data represent annual surveys from reefs with *A. palmata* ($n = 11$ reefs, 42 transects) and *A. cervicornis* ($n = 20$ reefs, 80 transects) in the Upper, Middle, and Lower Florida Keys (tables S5, S6, and S7). Yet, much of the *Acropora* decline on the FKRT occurred before this monitoring was established (15, 16). For example, there was a 96% decline in colony abundance of *A. cervicornis* at Molasses Reef between 1981 and 1986, and the live area coverage of *A. palmata* and *A. cervicornis* declined by 93 and 98% at Looe Key, respectively, from 1983 to 2000 (22, 26). Persistent recruitment failure and attrition of the remnant colonies as a result of disease, hurricanes, predation, and bleaching have all contributed to the lack of recovery (23, 25, 27, 28).

These severe declines led to the listing of these species as threatened under the Endangered Species Act in 2006, and both were elevated to critically endangered on the International Union for Conservation of Nature (IUCN) Red List in 2021 (9, 29). Rearing of *Acropora* spp. colonies in nurseries for restoration has increased over the past few decades (30), with a more coordinated effort recently enacted in the Florida Keys through the Mission: Iconic Reefs Program (31). The scope of restoration on FCR is evident: Many of the surveyed colonies (96.8%) were restored outplants (table S8). Before 2023, restoration increased coral cover fourfold and reestablished net positive carbonate production at some sites in Florida (32, 33). These restoration efforts may have prevented a full regional extinction of these species from all of FCR or at least preserved genetic diversity through field propagation, ex situ and in situ nurseries, and rescue to land-based holding facilities.

The historical importance of these two species cannot be overstated: For at least the past 250,000 to 500,000 years, *A. palmata* has dominated the reef crest zone, while *A. cervicornis* has dominated the mid-slope zone in the Caribbean (34). These species grow much faster than any other Atlantic corals, creating distinctive three-dimensional structural complexity unlike any other taxa (35). Of the two species, *A. palmata* has been the most important functionally because it was the primary reef builder in Florida during the Holocene (13, 17), is well adapted for high wave energy, and is the only scleractinian coral that both constructs and thrives within the reef crest habitat (34, 36). The architecture of both species supports economically important fisheries, and *A. palmata* is largely responsible for the shoreline protection provided by reefs (19).

The field surveys reported here were extensive, but it was impossible to survey every possible location where coral colonies may have been present; thus our surveys underestimate true population sizes. Although we observed 100% mortality in some regions (Middle and Lower Keys, Dry Tortugas), we cannot say with certainty whether or not these species are truly extinct in these places, because survivors may have persisted in locations that were not surveyed. Extinction events are difficult to document in the ocean owing to the challenges associated with surveying marine habitats (37). At a minimum, the 2023 marine heat wave caused a large number of local extinctions throughout the FKRT. Whether this can be classified as a regional extinction depends on how these regions are defined and on the possible presence of unknown survivors and their resilience potential.

Natural recovery of Florida's *Acropora* populations to the pre-1980s levels, the 1996 baseline (when sustained monitoring was established), or even the pre-2023 abundance levels is unlikely. Climate projections predict that annual severe bleaching ($\text{DHW} \geq 8$ °C-weeks per year) (38) will occur around 2040, whereas in situ warming trends

suggest that frequent bleaching episodes will begin sooner, perhaps as early as this decade (11). Neither projection, however, predicts the scale of extreme currently observed thermal anomalies. Recruitment failure in *A. palmata* was documented >20 years ago (27), as the loss of whole colonies, smaller colony size, and declining live-tissue area led to depressed fecundity and reproductive potential (39). The shift to fewer, smaller colonies with fewer branches has led to a loss of structural complexity, but it also limits the potential for asexual reproduction through fragmentation (27). The remnant *A. palmata* that survived the 2023 bleaching event are primarily small colonies located on a few reefs and remain vulnerable to disease, predation, storm damage, and future bleaching events. These colonies are unlikely to recover to a functional or reproductive size and may perish without intervention.

Because of northerly current trajectories, the surviving corals, which are mainly located in the far northern FCR, are unlikely to serve as sources of larvae to the FKRT (40). Chronic anthropogenic stress in this region and ongoing warming is likely to prevent these fossil reefs from serving as a thermal refuge or harboring self-sustaining populations of either species (17). Furthermore, the greatly reduced numbers of corals at sites where survivors exist lowers the odds of successful fertilization during spawning (41), because self-fertilization does not yield viable larvae in these species. Given the Caribbean-wide degradation of the acroporids and their low recruitment rates even before the dramatic recent declines (35, 42), ecologically important levels of replenishment from upstream populations (e.g., Yucatán) to the FKRT are unlikely in the near-term. The genetic connectivity of these species throughout the Caribbean (43), however, suggests that they could repopulate FCR over geological timescales, unless they experience a full extinction event in the Anthropocene.

The causal factors responsible for the long-term degradation of western Atlantic reefs are complex and debated, but contributing factors include overfishing of herbivorous fishes, land-based pollution, the mass die-off of the long-spined sea urchin (*Diadema antillarum*), the emergence of coral diseases, and coral bleaching (21, 44–46). Although the decline in Caribbean acroporids may have been underway as early as the 1950s (*A. palmata*) and 1960s (*A. cervicornis*) (46), it was in the 1980s that these reefs rapidly shifted from coral-dominated structures to those covered with macroalgae, sponges, and non-reef-building invertebrates (47). This macroalgae-dominated state exacerbated the already low rates of coral recruitment, resulting in little to no recovery after disturbance events (42). Almost all the wild *Acropora* populations have been lost from FCR, and the remnant reef structures are undergoing physical, biological, and chemical erosion, which could eventually completely eliminate this habitat and the ecosystem functions that these two species once supported.

Coral extirpation events are on the rise

Regardless of what initiated the demise of coral reefs in the wider Caribbean, or when it began, the 2023 marine heat wave has extirpated *Acropora* from most of the FKRT, completing the collapse of the two species on the reefs in this region. Although survivors persist in the northern portion of FCR, this population is unlikely to be self-sustaining owing to its low numbers, continued warming, and additional stressors (e.g., hurricanes and disease). As a result, *Acropora* are most likely functionally extinct throughout the entirety of FCR. This is not the first functional extinction of corals caused by coral bleaching: Two species of hydrocoral were extirpated from the eastern Pacific (37), and the pillar coral, *Dendrogyra cylindrus*, was also extirpated from FCR by 2020 by bleaching and stony coral tissue loss disease (48). The functional extinction of three coral species in Florida in <5 years is unprecedented. The hydrocoral *Millepora complanata* also became rare in Florida because of bleaching associated with the 1997–1998 El Niño (49). The functional and geological importance of these other taxa, however, pale in comparison with the two Atlantic acroporids:

A. palmata has been disproportionately responsible for Florida reef construction over the Holocene, as well as Caribbean reef construction for the past 250,000 to 500,000 years (13, 34). Thus, the 2023 bleaching and mortality event in Florida is not only ecologically but also geologically significant.

As the oceans continue to warm, temperatures may soon become inhospitable for many species, and true global extinctions could occur in the coming decades for the entire Caribbean biogeographic region. If marine heat waves with extreme heat anomalies become common, more extirpation events are likely to occur before the threat of annual severe bleaching is realized in 15 to 20 years. Future research is necessary to increase the temporal resolution of marine heat wave forecasts to include prediction of extreme thermal anomalies on daily-to-weekly timescales and to better understand the impacts of acute heat stress on coral reef and tropical taxa beyond reef-building corals. Aggressive interventions, such as introduction of heat-tolerant genotypes and genetic diversity from outside Florida, as well as algal symbiont manipulation, might be the only means to maintain any *Acropora* populations in Florida (50). Future restoration efforts that repeat what was done in the past are likely to result only in further losses. Restoration success, however, will likely be limited to localized areas and ultimately depend on the return time and severity of future bleaching events. Gene banking of the survivors and live biorepositories could be the only way to preserve the existing genetic richness of the acroporid corals that still remain in Florida and the Caribbean.

REFERENCES AND NOTES

1. S. Cooley et al., in *Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, H.-O. Pörtner et al., Eds. (Cambridge Univ. Press, 2022), pp. 379–550.
2. O. Hoegh-Guldberg et al., *Science* **382**, 1238–1240 (2023).
3. T. P. Hughes et al., *Nature* **556**, 492–496 (2018).
4. P. W. Glynn, *Coral Reefs* **12**, 1–17 (1993).
5. G. Liu et al., *Remote Sens.* **6**, 11579–11606 (2014).
6. N. E. Cantin, J. M. Lough, *PLOS ONE* **9**, e88720 (2014).
7. D. R. Levitan, W. Boudreau, J. Jara, N. Knowlton, *Mar. Ecol. Prog. Ser.* **515**, 1–10 (2014).
8. J. Miller et al., *Coral Reefs* **28**, 925–937 (2009).
9. L. Gutierrez et al., *PLOS ONE* **19**, e0309354 (2024).
10. J. T. Carlton, J. B. Geller, M. L. Reaka-Kudla, E. A. Norse, *Annu. Rev. Ecol. Syst.* **30**, 515–538 (1999).
11. D. P. Manzanillo, *Sci. Rep.* **5**, 16762 (2015).
12. J. W. Porter, S. K. Lewis, K. G. Porter, *Limnol. Oceanogr.* **44**, 941–949 (1999).
13. L. T. Toth et al., *Sci. Rep.* **11**, 13044 (2021).
14. E. A. Shinn, *J. Sediment. Petrol.* **33**, 291–303 (1963).
15. J. W. Porter, O. W. Meier, *Am. Zool.* **32**, 625–640 (1992).
16. E. A. Shinn, I. B. Kuffner, Florida Keys Corals: A Photographic Record of Changes from 1959 to 2015 (USGS Data Release, 2017); <https://doi.org/10.5066/F7S46QWR>.
17. A. B. Modys, A. E. Oleinik, L. T. Toth, W. F. Precht, R. A. Mortlock, *Commun. Earth Environ.* **5**, 119 (2024).
18. A. Stathakopoulos, L. T. Toth, A. B. Modys, S. A. Johnson, I. B. Kuffner, *Depos. Rec.* **11**, 808–828 (2025).
19. I. B. Kuffner, L. T. Toth, *Conserv. Biol.* **30**, 706–715 (2016).
20. J. B. C. Jackson, M. K. Donovan, K. L. Cramer, V. Y. Y. Lam, Status and trends of Caribbean coral reefs: 1970–2012 (Global Coral Reef Monitoring Network, IUCN, 2014); https://iucn.org/sites/default/files/import/downloads/caribbean_coral_reefs_status_report_1970_2012.pdf.
21. R. B. Aronson, W. F. Precht, *Hydrobiologia* **460**, 25–38 (2001).
22. M. W. Miller, A. S. Bourque, J. A. Bohnsack, *Coral Reefs* **21**, 179–182 (2002).
23. M. W. Miller et al., “*Acropora* corals in Florida: Status, trends, conservation, and prospects for recovery” in *Proceedings of the Caribbean Acropora workshop: Potential application of the US Endangered Species Act as a conservation strategy*, Miami, FL, USA, 16 to 18 April 2002, A. W. Bruckner, Ed. (NOAA Technical Memorandum NMFS-OPR-24, 2002), pp. 59–70.
24. D. E. Williams, M. W. Miller, *Coral Reefs* **31**, 369–382 (2012).
25. D. E. Williams, M. W. Miller, A. J. Bright, R. E. Pausch, A. Valdivia, *Mar. Pollut. Bull.* **124**, 189–197 (2017).
26. W. C. Jaap, J. C. Halas, R. G. Muller, in *Proceedings of the 6th International Coral Reef Symposium*, Townsville, Australia, 8 to 12 August 1988, J. H. Choat et al., Eds. (James Cook Univ. Press, 1988), pp. 237–243.
27. D. E. Williams, M. W. Miller, K. L. Kramer, *Coral Reefs* **27**, 697–705 (2008).

28. L. M. Harper *et al.*, *Mar. Ecol. Prog. Ser.* **721**, 39–58 (2023).
29. National Oceanic and Atmospheric Administration (NOAA), National Marine Fisheries Service, “Endangered and threatened species: Final listing determinations for elkhorn coral and staghorn coral” (*Federal Register*, vol. 71, 2006), pp. 26852–26861.
30. C. N. Young, S. A. Schopmeyer, D. Lirman, *Bull. Mar. Sci.* **88**, 1075–1098 (2012).
31. National Oceanic and Atmospheric Administration (NOAA), Restoring Seven Iconic Reefs: A Mission to Recover the Coral Reefs of the Florida Keys (2022); https://media.fisheries.noaa.gov/dam-migration/restoring_seven_iconic_reefs_-_a_mission_to_recover_the_coral_reefs_of_the_florida_keys.pdf.
32. M. C. Ladd, D. E. Burkepile, A. A. Shantz, *Restor. Ecol.* **27**, 1166–1176 (2019).
33. E. Esplandiú, J. Morris, I. Enochs, N. Besemer, D. Lirman, *Sci. Rep.* **14**, 27599 (2024).
34. J. M. Pandolfi, J. B. C. Jackson, *Ecol. Lett.* **9**, 818–826 (2006).
35. C. T. Perry *et al.*, *Nat. Commun.* **4**, 1402 (2013).
36. T. F. Goreau, *Ecology* **40**, 67–90 (1959).
37. P. W. Glynn, *Bull. Mar. Sci.* **87**, 767–794 (2011).
38. R. van Hooidonk *et al.*, *Projections of Future Coral Bleaching Conditions Using IPCC CMIP6 Models: Climate Policy Implications, Management Applications, and Regional Seas Summaries* (United Nations Environment Programme, 2020).
39. D. E. Williams, M. W. Miller, I. B. Baums, *Coral Reefs* **33**, 595–606 (2014).
40. C. Drury, C. B. Paris, V. H. Kourafalou, D. Lirman, *Coral Reefs* **37**, 585–596 (2018).
41. P. J. Mumby *et al.*, *Proc. Natl. Acad. Sci. U.S.A.* **121**, e2418314121 (2024).
42. G. Roff, *Bioscience* **71**, 26–39 (2021).
43. I. B. Baums, M. W. Miller, M. E. Hellberg, *Mol. Ecol.* **14**, 1377–1390 (2005).
44. H. A. Lessios, *Annu. Rev. Ecol. Syst.* **19**, 371–393 (1988).
45. J. B. C. Jackson *et al.*, *Science* **293**, 629–637 (2001).
46. K. L. Cramer *et al.*, *Sci. Adv.* **6**, eaax9395 (2020).
47. T. P. Hughes, *Science* **265**, 1547–1551 (1994).
48. K. L. Neely, C. L. Lewis, K. S. Lunz, L. Kabay, *Front. Mar. Sci.* **8**, 656515 (2021).
49. R. R. Ruzicka *et al.*, *Mar. Ecol. Prog. Ser.* **489**, 125–141 (2013).
50. R. F. Karp *et al.*, *Coral Reefs* **44**, 953–965 (2025).
51. R. Cunnig, R. F. Karp, Data and code for: Heat-driven functional extinction of Caribbean *Acropora* corals from Florida’s Coral Reef, GitHub (2025); <https://github.com/jrcunning/FLAcro23>.
52. R. Cunnig, R. F. Karp, Data and code for: Heat-driven functional extinction of Caribbean *Acropora* corals from Florida’s Coral Reef, Zenodo (2025); <https://doi.org/10.5281/zenodo.15701619>.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S5; Tables S1 to S8; References (53–66); MDAR Reproducibility Checklist

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Heat-driven functional extinction of Caribbean *Acropora* corals from Florida's Coral Reef

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Editor's summary

As little as a 2°C increase in temperature is enough for many corals to bleach, a process in which symbionts are shed and from which recovery is uncertain. An increase of more than a couple of degrees kills corals, and their reef-building skeletons start to crumble. Manzello *et al.* have been tracking the fate of species of *Acropora* corals along the Florida Keys. These species have been struggling to survive elevated temperatures for many years, and there have been intensive conservation efforts to restore *Acropora* spp. reefs. Unfortunately, in 2023, an acute heating event, when water temperatures exceeded the maximum monthly mean sea surface temperature by 2.5°C for many weeks, led to the extinction of wild and replanted corals on the Florida Keys. —Caroline Ash

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